CHAPTER

03

Learning and Theories of Learning

Imagine waking up having lost everything you ever learned: you could not read, write, speak, feed yourself or find your way home. Almost all human behaviour is **learned**. This chapter explains what learning is (and is not), what helps or hinders it, and the four major theories that explain how it happens: **classical conditioning** (Pavlov), **operant conditioning** (Skinner), **social learning** (Bandura) and **cognitive learning** (latent and insight learning). It is one of the most heavily examined chapters, especially through real-life scenario questions.

LESSONS IN THIS CHAPTER

3.1	The Nature of Learning
3.2	Classical Conditioning (Pavlov)
3.3	Operant Conditioning: Reinforcement and Punishment
3.4	Schedules of Reinforcement and Shaping
3.5	Social Learning and Cognitive Learning

Freshman A+ · General Psychology (Psyc 1011)

CHAPTER 3 · LESSON 3.1

The Nature of Learning



AFTER THIS LESSON YOU CAN

- Define learning and explain each part of the definition
- Distinguish learning from maturation, fatigue, illness and drugs
- List the characteristics and principles of learning
- Explain the factors that influence learning

3.1.1 Definition of learning

D LEARNING

Learning is a relatively permanent change in behaviour that occurs as a result of experience or practice.

F WHAT THE DEFINITION INCLUDES AND EXCLUDES

Part of the definition	Meaning	So this is NOT learning
a change in behaviour	for **better or worse** (bad habits are learned too)	
relatively permanent	lasting, not momentary	changes due to **fatigue**, **illness**, **drugs/alcohol** (temporary)
from experience or practice	the learner **interacts with the environment**	changes due to **maturation**, **growth**, **hormones**, **reflexes** (biological)
not directly observable	learning itself is **inferred** from changes in performance	

LEARNING OR NOT?

- A child bitten by a dog avoids dogs for years → **learning** (experience, lasting).
- A student who performs poorly because she is tired → **not** learning (fatigue, temporary).
- A boy's voice deepens at puberty → **not** learning (hormones / maturation).
- A baby sucks at the breast from birth → **not** learning (an inborn reflex).
- A child learns to speak her mother tongue → **learning**.
- A student practises the piano for months and becomes skilled → **learning** (relatively permanent change through practice).

3.1.2 Characteristics of learning (Yoakman and Simpson)

1. **Continuous** modification of behaviour **throughout life**.
2. **Pervasive:** it reaches into **all aspects** of human life.
3. Involves the **whole person**: socially, emotionally and intellectually.
4. Often a change in the **organization of experience**.
5. **Responsive to incentives**.
6. An **active** process (**not passive**).
7. **Purposeful**.
8. Depends on **maturation, motivation and practice**.
9. **Multifaceted**.

3.1.3 Principles of learning

Principle	Meaning
Readiness	we learn best when physically, mentally and emotionally ready
Exercise (practice)	meaningful practice improves learning and retention
Effect	learning is strengthened by **pleasant** feelings and weakened by unpleasant ones
Primacy	things learned **first** create a strong impression
Recency	things learned **most recently** are best remembered
Intensity	we learn more from the **real thing** than from a substitute
Freedom	things freely learned are best learned

3.1.4 Factors influencing learning

Motivation (stronger, clearer motives → more effort); **maturation** (neuromuscular readiness); **health** (sensory defects, malnutrition, lack of sleep and fatigue hinder learning); **psychological well-being** (worry, fear, loneliness and inferiority hinder it; self-confidence helps); **working conditions** (fresh air, light, comfort, no noise); **background experience** (relating new material to what you already know); **length of study period** (neither too long nor too short); **massed vs distributed practice** (spreading study over time with breaks works better than cramming).

! NATURE-OF-LEARNING TRAPS

- "Learning can be observed directly" is **false**; we infer it from behaviour.
- "Learning is passive" is **false**; it is active.
- Learning includes changes for the **worse** as well as for the better.
- Changes due to **growth and maturation** are **not** learning, even if they are permanent.
- "Pervasive" means it reaches into **all aspects** of life.

Exam Practice

11 QUESTIONS · LESSON 3.1

1 Addis Ababa University · Mid 2014

Regarding the characteristics of learning, which of the following is NOT true? Learning is:

A) passive
B) pervasive
C) purposeful
D) multifaceted

2 Addis Ababa University · Mid 2014

Which of the following is NOT true about learning? Learning:

A) is a change in behaviour
B) depends on practice
C) can be observed directly
D) results in an enduring change

3 Bahir Dar University · Final 2015

All of the following describe learning EXCEPT:

A) It is a change in behaviour for better or worse
B) It can occur through practice or experience
C) It includes changes due to growth and maturation
D) It is relatively permanent

4 Dilla University · Mid 2015

"Changes that result from fatigue and maturation cannot be considered learning." This shows that:

A) learning is a process
B) changes in learning should be relatively permanent and come from experience
C) learning depends on practice
D) learning is always observable

5 Arba Minch University · Final 2015

In the definition of learning, the phrase "result of practice or experience" implies that:

A) a newly learnt behaviour should last forever
B) a learner has to interact with the environment to learn new behaviour
C) learning takes place only inside the mind
D) learning cannot change

6 Bahir Dar University · Mid 2014

A baby's breast-sucking behaviour is the result of learning.

True / False

7 Haramaya University · Mid 2015

Learning is a change in behaviour that includes both positive and negative changes.

True / False

8 Arba Minch University · Final 2015

Which of the following is FALSE about the principles of learning?

A) Substitute objects produce more intense learning than real ones
B) Learning is strengthened when followed by pleasant consequences
C) Things learned first produce strong impressions
D) Recently learned things are remembered best

9 Jimma University · Mid 2015

"Students learn best and retain information longer when they have meaningful practice and exercise." This principle is:

A) Readiness
B) Practice (exercise)
C) Effect
D) Primacy

10 Jimma University · Mid 2015

"All related facts and understandings from a previously learned course should be brought to new learning." Which factor affecting learning is this?

A) Background experience
B) Motivation
C) Maturation
D) Psychological well-being

11 Madda Walabu University · Final 2014

What does "learning is pervasive" mean?

A) It is a continuous modification
B) It is purposeful
C) It reaches into all aspects of life
D) It is active

Answers & Explanations

1. A

passive

Learning is an active process.

2. C

can be observed directly

Learning is inferred from changes in performance; it cannot be seen directly.

3. C

It includes changes due to growth and maturation

Growth and maturation are biological changes, not learning.

4. B

changes in learning should be relatively permanent and come from experience

Fatigue is temporary and maturation is biological; learning must be a relatively permanent change caused by experience.

5. B

a learner has to interact with the environment to learn new behaviour

Experience and practice come from interacting with the environment, which distinguishes learning from biological change.

6. False

Sucking is an inborn reflex, present without experience.

7. True

Bad habits such as smoking are learned just like good ones.

8. A

Substitute objects produce more intense learning than real ones

The principle of intensity says we learn more from the real thing than from a substitute.

9. B

Practice (exercise)

It describes the principle of exercise (practice).

10. A

Background experience

Linking new learning to prior knowledge is the factor of background experience.

11. C

It reaches into all aspects of life

Pervasive means spreading into every area of human life.

CHAPTER 3 · LESSON 3.2

Classical Conditioning (Pavlov)

AFTER THIS LESSON YOU CAN

- Explain classical conditioning and Pavlov's experiment
- Identify NS, UCS, UCR, CS and CR in any scenario
- Explain generalization, discrimination, extinction and spontaneous recovery
- Apply classical conditioning to fears and everyday reactions

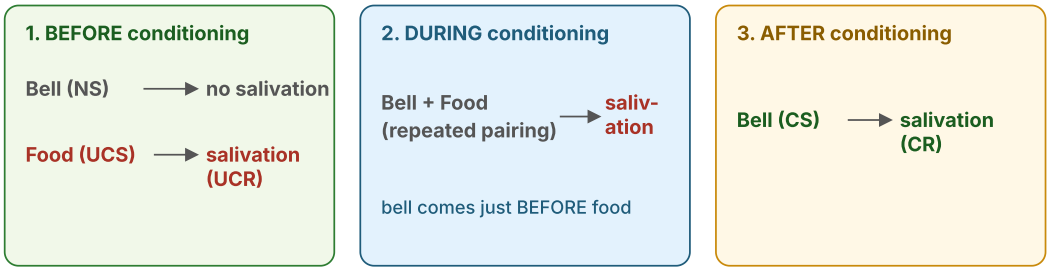
3.2.1 What classical conditioning is

D CLASSICAL CONDITIONING

A type of learning in which a **neutral stimulus**, after being **paired** with a stimulus that **naturally** produces a response, comes to produce a similar response by itself. It deals with **involuntary** emotional and physiological responses (fear, salivation, nausea, increased heartbeat). It is also called **substitution learning**, because the neutral stimulus is substituted for the natural stimulus. Discovered by the Russian physiologist **Ivan Pavlov**.

F THE FIVE TERMS

Term	Meaning	In Pavlov's experiment
Neutral stimulus (NS)	produces **no** response of interest **before** conditioning	the bell (at first)
Unconditioned stimulus (UCS)	**naturally** produces a response, no learning needed	food (meat)
Unconditioned response (UCR)	the **natural**, unlearned response to the UCS	salivation to food
Conditioned stimulus (CS)	the former NS, **after** pairing with the UCS	the bell (after training)
Conditioned response (CR)	the **learned** response to the CS	salivation to the bell



Pavlov's three stages of classical conditioning

For the strongest conditioning, the **NS should come just before** the UCS (a buzzer immediately followed by food), and the pairing should be **repeated**.

? SOLVING A CLASSICAL CONDITIONING SCENARIO (JIMMA 2015 / DILLA 2015)

"Chala's mother buys him a sailor's cap before a family boat trip. On the boat, Chala gets seasick and vomits. The next day he feels sick just looking at the cap." Identify every term.

Step 1: Find the **natural** cause–effect pair (no learning needed): boat motion → nausea. So **UCS = boat motion**, **UCR = nausea on the boat**.

Step 2: Find the thing that was **neutral** but present at the same time: the **cap (NS)**.

Step 3: After pairing, the cap alone causes nausea: **CS = the cap**, **CR = nausea at the sight of the cap**.

Rule: the UCS and UCR are the "**natural**" pair; the CS is the thing that **used to be neutral**; the CR is usually the **same kind of reaction as the UCR**, now produced by the CS.

MORE REAL-LIFE CLASSICAL CONDITIONING

- **Cancer patients** become nauseated when they approach the hospital: UCS = chemotherapy; UCR = nausea; CS = the building; CR = nausea at the building.
- **Mamo** loves bananas; his father always brings bananas in a bag. Now Mamo smiles whenever he sees his father carrying a bag: UCS = banana; UCR = happiness at the banana; CS = the bag; CR = smiling at the bag.
- A **child shocked by an electrical outlet** while playing with a toy now fears the toy: UCS = shock; UCR = fear/crying; CS = the toy; CR = fear of the toy.
- Watson's **Little Albert:** a loud noise (UCS) made a little boy cry (UCR); after the noise was paired with a white rat, the rat (CS) alone made him cry (CR).

3.2.2 Principles of classical conditioning

FOUR PRINCIPLES

Principle	What happens	Example
Stimulus generalization	stimuli **similar** to the CS also produce the CR	bitten by a scorpion → now fears spiders too; after malaria, fears **all** insects; a baby calls **all** men "daddy"
Stimulus discrimination	responding to the CS but **not** to similar stimuli	Alemu fears riding a **motorbike** after an accident, but not a **bicycle**; the dog salivates to the dinner bell, not the doorbell
Extinction	the **CS is presented repeatedly without the UCS**, so the CR weakens and disappears	ring the bell many times with no food → salivation stops
Spontaneous recovery	after extinction and a **rest period**, the CR **suddenly returns** (weaker)	a man who quit smoking months ago suddenly craves a cigarette when he sees someone smoking

EXTINCTION IS NOT FORGETTING

Extinction only **inhibits** the CR; it does not erase it. That is why **spontaneous recovery** can happen. The recovered CR is weaker and extinguishes faster.

CLASSICAL CONDITIONING TRAPS

- Extinction = **CS without UCS** (not UCS without CS).
- After conditioning, the **natural stimulus does not become a learned stimulus**; the **neutral** stimulus becomes the CS.
- The **new, learned behaviour** is the **CR**.
- Classical conditioning deals with **involuntary** responses; operant conditioning deals with **voluntary** ones. Statements that reverse this are false.
- Applying lab safety rules learned in chemistry to the biology lab = **generalization**, not discrimination.

Exam Practice

20 QUESTIONS · LESSON 3.2

1 Arsi University · Final 2013

The previously neutral stimulus that, after conditioning, can evoke a response is the:

A) unconditioned response
B) conditioned response
C) conditioned stimulus
D) unconditioned stimulus

2 Arsi University · Final 2013

In classical conditioning, a stimulus that produces a response without previous training is the:

A) unconditioned stimulus
B) conditioned stimulus
C) conditioned response
D) unconditioned response

3 Bahir Dar University · Final 2015

According to classical conditioning theory, the new (learned) behaviour is:

A) the unconditioned stimulus
B) the conditioned stimulus
C) the conditioned response
D) the unconditioned response

4 Dilla University · Mid 2015

Tedla went on a boat trip on Lake Hawassa, became nauseated and vomited. The next day he felt nauseated when he saw a boat on the shore. The boat has become a(n):

A) conditioned stimulus
B) conditioned response
C) unconditioned stimulus
D) unconditioned response

5 Jimma University · Mid 2015

Many cancer patients who have repeated chemotherapy become nauseated when they approach, or even imagine, the hospital building. The building and the chemotherapy are, respectively, the:

A) unconditioned stimulus and conditioned stimulus
B) conditioned response and unconditioned response
C) conditioned stimulus and unconditioned stimulus
D) unconditioned stimulus and neutral stimulus

6 Jigjiga University · Mid 2012

Abdu becomes happy after eating chocolate. His English teacher often offers him chocolate, and now he feels happy just seeing the teacher. The teacher is a(n):

A) unconditioned stimulus
B) unconditioned response
C) conditioned stimulus
D) conditioned response

7 Haramaya University · Mid 2015

Children salivate when they see their teacher wash her hands before serving food. The hand-washing represents:

A) Extinction
B) Unconditioned response
C) Conditioned response
D) Conditioned stimulus

8 Jigjiga University · Mid 2014

Mrs. Zeynab often chopped onion and potato together, and the onion made her cry. Today she forgot the onion and chopped only potato, yet tears came. The conditioned stimulus is:

A) the onion
B) the potato
C) the tears
D) none of these

9 Addis Ababa University · Mid 2014

After Debela was conditioned to fear a black lion, he also showed fear of a black dog. This is an example of:

A) stimulus discrimination
B) stimulus generalization
C) spontaneous recovery
D) acquisition

10 Bahir Dar University · Final 2015

A six-year-old who was stung by a scorpion learned to fear scorpions and then began to fear spiders. This is best described as:

A) Spontaneous recovery
B) Stimulus discrimination
C) Extinction
D) Stimulus generalization

11 Bahir Dar University · Mid 2014

Alemu, recovering from a serious motorbike accident, is afraid to ride a motorbike but not a bicycle. His fear illustrates:

A) Stimulus generalization
B) Spontaneous recovery
C) Stimulus discrimination
D) Extinction

12 Wachemo University · Mid 2016

A child has learned to avoid a furry black cat, but she still plays with her grandmother's short-haired tabby. Her response demonstrates:

A) Extinction
B) Stimulus generalization
C) Spontaneous recovery
D) Stimulus discrimination

13 Bahir Dar University · Final 2015

A 30-year-old man stopped smoking several months ago, but the sight of someone else smoking suddenly makes him want to smoke. According to classical conditioning, this is:

A) Extinction
B) Stimulus generalization
C) Stimulus discrimination
D) Spontaneous recovery

14 Haramaya University · Mid 2015

In classical conditioning, the procedure for extinguishing a response is to:

A) present the unconditioned stimulus without the conditioned stimulus
B) present the conditioned stimulus without the unconditioned stimulus
C) punish the response
D) increase the number of pairings

15 Debre Birhan University · Final 2015

During extinction, the **__ is presented alone and the ____** gradually decreases.

A) CS; CR
B) UCS; CR
C) UCS; UCR
D) CS; UCR

16 Arba Minch University · Final 2016

In Pavlov's experiment, which pairing would be most effective in establishing a conditioned response?

A) A loud buzzer immediately followed by food
B) A soft tone presented several seconds after the food
C) A flashing light presented repeatedly without food
D) Food presented without any other stimulus

17 Arsi University · Final 2013

If a student applies the safety rules of the chemistry lab to the biology lab, this is stimulus discrimination.

True / False

18 Arba Minch University · Final 2016

After conditioning, the natural stimulus becomes the learned stimulus.

True / False

19 Bahir Dar University · Mid 2014

Whenever a girl visits her uncle's house, a dog's barking frightens her; as a result she develops fear of her uncle's house. Identify the UCS, UCR, CS and CR.

20 Model question (module activity)

A one-year-old pushes part of his toy into an electrical outlet, gets a shock, becomes frightened and cries. For days afterwards he shows fear when his mother gives him the toy. Identify the UCS, UCR, CS and CR, and show the three stages.

Answers & Explanations

1. C
conditioned stimulus
Through pairing with the UCS, the neutral stimulus becomes the conditioned stimulus.

2. A
unconditioned stimulus
The UCS (e.g. food) naturally produces the UCR.

3. C
the conditioned response
The CR is the learned response to the former neutral stimulus.

4. A
conditioned stimulus
The boat was neutral before; after pairing with the motion sickness, it triggers nausea by itself.

5. C
conditioned stimulus and unconditioned stimulus
Chemotherapy naturally causes nausea (UCS); the building has been paired with it (CS).

6. C

conditioned stimulus

Chocolate is the UCS; the teacher, repeatedly paired with chocolate, has become the CS.

7. D

Conditioned stimulus

Hand-washing was neutral but has been paired with food, so it now triggers salivation.

8. B

the potato

The onion naturally causes tears (UCS); the potato, paired with it, became the CS. The tears to the potato are the CR.

9. B

stimulus generalization

A stimulus similar to the CS (black animal) produces the same CR.

10. D

Stimulus generalization

The fear spread to a similar stimulus.

11. C

Stimulus discrimination

He responds to the CS (motorbike) but not to the similar stimulus (bicycle).

12. D

Stimulus discrimination

She distinguishes between the feared cat and a different-looking cat.

13. D

Spontaneous recovery

An extinguished response reappears after a period of rest.

14. B

present the conditioned stimulus without the unconditioned stimulus

The CS repeatedly presented alone loses its power to produce the CR.

15. A

CS; CR

The CS is presented without the UCS, so the CR weakens.

16. A

A loud buzzer immediately followed by food

Conditioning is strongest when the neutral stimulus comes just before the UCS. C would produce no conditioning (or extinction).

17. False

Transferring a response to a similar situation is stimulus generalization.

18. False

The **neutral** stimulus becomes the conditioned (learned) stimulus; the natural stimulus remains the UCS.

19. **UCS = the dog's barking; UCR = fear (frustration) at the barking; CS = the uncle's house; CR = fear of the uncle's house**

The barking naturally frightens her; the house, repeatedly paired with the barking, now produces fear by itself.

20. **UCS = electric shock; UCR = fear and crying from the shock; CS = the toy; CR = fear of the toy**

Before: toy → no fear; shock → fear. During: toy + shock → fear. After: toy alone → fear.

CHAPTER 3 · LESSON 3.3

Operant Conditioning: Reinforcement and Punishment

AFTER THIS LESSON YOU CAN

- Explain operant conditioning and Skinner's view
- Distinguish positive and negative reinforcement and punishment
- Distinguish primary and secondary reinforcers and punishers
- Explain escape and avoidance learning and the problems with punishment

3.3.1 What operant conditioning is

D **OPERANT (INSTRUMENTAL) CONDITIONING**

Learning in which a **voluntary** response is **strengthened or weakened** by its **consequences**. The organism **operates** on its environment, and the consequence influences whether the response will occur again. The **response comes before** the consequence. Main proponent: **B. F. Skinner**, who said that to explain behaviour we should look **outside** the person, at the consequences.

	Classical conditioning	Operant conditioning
Founder	Pavlov	Skinner (building on Thorndike's law of effect)
Type of response	**involuntary** (reflexes, emotions)	**voluntary** (deliberate actions)
Learning by	association of two **stimuli**	association of a **behaviour** with its **consequence**
Order	stimulus → response	response → consequence
Similarity	both are **behavioural** theories: the **environment** shapes learning through **stimulus–response associations**	

3.3.2 Consequences: reinforcement and punishment

A response can have three kinds of consequence: **neutral** (no change), **reinforcement** (makes it **more** likely) or **punishment** (makes it **less** likely).

THE 2 × 2 TABLE: MEMORISE IT

	Add (+) a stimulus	Remove (−) a stimulus
Behaviour increases (reinforcement)	**Positive reinforcement**: add something **pleasant** (praise, money, good marks)	**Negative reinforcement**: remove something **unpleasant** (nagging stops, headache goes, noise stops)
Behaviour decreases (punishment)	**Positive punishment**: add something **unpleasant** (extra work, insults, pinching, a fine notice)	**Negative punishment**: remove something **pleasant** (taking away medals, privileges, grades, game time)
Positive = add. Negative = remove. Reinforcement = increase. Punishment = decrease. "Negative" does not mean "bad".		

THE TWO-QUESTION METHOD (DEBRE BIRHAN 2015)

"A child cries continuously while his mother is at work. When she picks him up, he stops crying. Which consequence is acting on the mother?"

Question 1: **Whose behaviour** are we analysing, and does it **increase** or **decrease**? The mother's behaviour (picking up the child) will **increase** → reinforcement.

Question 2: Was something **added** or **removed**? The unpleasant crying was **removed** → **negative reinforcement** for the mother.

(For the child, crying was followed by being picked up, something pleasant added → **positive reinforcement** for crying.)

CLASSIFY EACH CONSEQUENCE

- A teacher loads **extra work** on a misbehaving student → **positive punishment** (unpleasant added).
- An athlete's medals are **taken away** after a doping test → **negative punishment** (pleasant removed).
- A player is **benched** for poor play → **negative punishment**.
- An instructor changes a cheater's grade from **A to F** → **negative punishment**.
- Dorm-mates **insult** a messy student → **positive punishment**.
- Lomi has a headache, goes for a run and the **headache disappears**; next time she wants to run → **negative reinforcement**.
- Wearing a **face mask** to avoid COVID-19 → **negative reinforcement** (avoiding something unpleasant).
- Taking **aspirin** to remove a headache → **negative reinforcement**.
- A mother warns that her son must wash the dishes on Saturday **unless** he cleans his room → he cleans to **avoid** the chore → **negative reinforcement**.

ESCAPE AND AVOIDANCE LEARNING

- **Escape learning:** learning a response that **stops** an unpleasant stimulus that is already present. A teacher **closes the door** because students' noise outside is disturbing him.
- **Avoidance learning:** learning to **prevent** an unpleasant stimulus **before** it occurs (wearing a mask before you are infected).

  Both are maintained by **negative reinforcement**.

3.3.3 Primary and secondary reinforcers

	Reinforcers	Punishers
Primary (natural, satisfy biological needs, no learning needed)	food, water, warmth, comfortable air temperature, sex, touch	pain, extreme heat or cold
Secondary (gain power through **association** with primary ones)	**money**, praise, applause, good grades, awards, gold stars	criticism, scolding, catcalls, fines, demerits, bad grades

3.3.4 Effective punishment and its problems

Punishment works best when it is **immediate**, **consistent** and of sufficient **intensity**. But it often fails because:

1. it is given **inappropriately**, in anger, and covers irrelevant behaviour;
2. it produces **anxiety, fear or rage**, which generalize (by classical conditioning) to the person and place: a student may come to hate the teacher, the subject and school; harsh physical punishment is a risk factor for depression, low self-esteem and violence;
3. its effect is often **temporary**, lasting only while the punisher is present;
4. most behaviour is hard to punish **immediately**;
5. it gives **little information** about what to do instead, and the attention it brings may even **reinforce** the behaviour.

! OPERANT TRAPS

- **Negative reinforcement increases** behaviour; it is **not** punishment.
- "Both negative reinforcement and negative punishment decrease behaviour" is **false**.
- Pinching a student is **positive** punishment (something added), not negative.
- **Money** and **grades** are **secondary** reinforcers, not primary.
- Both **positive and negative reinforcement increase** behaviour; both **positive and negative punishment decrease** it.

Exam Practice

23 QUESTIONS · LESSON 3.3

1 Arsi University · Final 2013

Positive reinforcement __ **behaviour, and negative reinforcement** ____ behaviour.

A) strengthens; strengthens
B) weakens; weakens
C) strengthens; weakens
D) weakens; strengthens

2 Jimma University · Mid 2015

Positive punishment __ **behaviour and negative punishment** ____ behaviour.

A) strengthens; strengthens
B) weakens; weakens
C) strengthens; weakens
D) weakens; strengthens

3 Arsi University · Final 2013

Negative reinforcement decreases behaviour and makes it less likely to recur.

True / False

4 Arsi University · Final 2014

Both negative reinforcement and negative punishment decrease the likelihood of a behaviour.

True / False

5 Arsi University · Final 2013

A teacher who pinches students who come to class without a face mask is practising negative punishment.

True / False

6 Bahir Dar University · Final 2015

Negative punishment is:

A) removal of an unpleasant stimulus
B) removal of a pleasant stimulus
C) giving an aversive stimulus
D) presenting a pleasant stimulus

7 ASTU · Mid 2015

World Athletics takes away all the medals won by athlete X after confirming that the medals were won with the help of prohibited drugs. What kind of consequence is this?

A) Positive punishment
B) Negative punishment
C) Positive reinforcement
D) Negative reinforcement

8 Arba Minch University · Final 2015

A player who is benched for not performing well during a match is experiencing:

A) Negative punishment
B) Negative reinforcement
C) Positive punishment
D) Positive reinforcement

9 Bahir Dar University · Mid 2014

Loading extra work on a misbehaving student is best explained by:

A) Positive punishment
B) Positive reinforcement
C) Negative punishment
D) Negative reinforcement

10 Mekelle University · Mid 2011

Mahider's dorm room is a mess because she studies all the time and is reluctant to clean. Her dorm-mates now insult her to make her clean. Her dorm-mates are using:

A) Positive reinforcement
B) Positive punishment
C) Negative punishment
D) Negative reinforcement

11 Jimma University · Mid 2015

When a child misbehaves, a caregiver removes him from a desirable activity to decrease the unwanted behaviour. The caregiver is using:

A) Positive reinforcement
B) Positive punishment
C) Negative reinforcement
D) Negative punishment

12 Debre Birhan University · Final 2015

While his mother is at work, a child cries continuously; when she picks him up, he stops. From this case, the mother is:

A) positively reinforced
B) negatively reinforced
C) positively punished
D) negatively punished

13 Madda Walabu University · Final 2014

Lomi has a splitting headache. She goes for a run and the headache disappears. The next time she has a headache, she thinks of running. The disappearance of the headache is:

A) positive reinforcement
B) negative reinforcement
C) negative punishment
D) positive punishment

14 Bahir Dar University · Mid 2014

Using a face mask to avoid getting infected with COVID-19 is a good example of:

A) Positive reinforcement
B) Negative punishment
C) Positive punishment
D) Negative reinforcement

15 Dire Dawa University · Mid 2014

Which of the following is the best example of negative reinforcement?

A) A child being spanked for bad behaviour
B) A kindergarten student put in "time-out"
C) A teenager not allowed to go to her friend's party
D) A mother taking aspirin to eliminate her headache

16 Madda Walabu University · Final 2014

A teacher closes the door because the noise of students outside disturbs him. The teacher is showing:

A) Escape learning
B) Avoidance learning
C) Insight learning
D) Latent learning

17 Jimma University · Final 2014

A mother tells her son that he will have to wash the dishes on Saturday if he does not clean his room by the end of the week. The mother is using:

A) Positive reinforcement
B) Shaping
C) Negative reinforcement
D) Negative punishment

18 Madda Walabu University · Final 2014

Which of the following is NOT a primary reinforcer?

A) Food
B) Money
C) Water
D) Sex

19 Arba Minch University · Final 2015

Grades and praise from teachers serve as primary reinforcers.

True / False

20 Arsi University · Final 2013

Secondary reinforcers gain their reinforcing qualities through association with primary reinforcers.

True / False

21 Arsi University · Final 2014

Identify the odd one out from a punisher point of view.

A) Slapping
B) Pinching
C) Criticism
D) Burning heat

22 Bahir Dar University · Mid 2014

In which assumption are classical and operant conditioning similar?

A) The impact of the unconditioned stimulus
B) The effect of punishment and reinforcement
C) The impact of the environment on learners' behaviour
D) Voluntary responses of the learner

23 ASTU · Mid 2015

Unlike classical conditioning, which depends on voluntary responses, operant conditioning applies to involuntary responses.

True / False

Answers & Explanations

1. A

strengthens; strengthens

All reinforcement strengthens behaviour; "negative" only means something unpleasant is removed.

2. B

weakens; weakens

All punishment weakens behaviour.

3. False

Negative reinforcement increases behaviour by removing an unpleasant stimulus.

4. False

Negative reinforcement increases behaviour; only negative punishment decreases it.

5. False

Pinching adds an unpleasant stimulus: positive punishment.

6. B

removal of a pleasant stimulus

Taking away something pleasant to decrease behaviour is negative punishment.

7. B

Negative punishment

Something pleasant (the medals) is removed to discourage doping.

8. A

Negative punishment

Playing time (pleasant) is taken away.

9. A

Positive punishment

An unpleasant task is added to decrease misbehaviour.

10. B

Positive punishment

Insults are an unpleasant stimulus added after the unwanted behaviour (not cleaning).

11. D

Negative punishment

A pleasant activity is taken away ("time-out").

12. B

negatively reinforced

Her picking-up behaviour is strengthened because it removes the unpleasant crying.

13. B

negative reinforcement

Removal of an unpleasant stimulus (headache) strengthens running.

14. D

Negative reinforcement

Mask-wearing is strengthened because it prevents an unpleasant outcome (avoidance learning).

15. D

A mother taking aspirin to eliminate her headache

Taking aspirin is strengthened by the removal of pain. A = positive punishment; B and C = negative punishment.

16. A

Escape learning

He makes a response that stops an unpleasant stimulus that is already present.

17. C

Negative reinforcement

Cleaning the room is strengthened because it lets the boy avoid an unpleasant chore.

18. B

Money

Money is secondary: its value is learned through association with primary reinforcers.

19. False

They are secondary reinforcers.

20. True

Money is valuable because it buys food, shelter and comfort.

21. C

Criticism

Criticism is a secondary (learned) punisher; the others cause physical pain, so they are primary punishers.

22. C

The impact of the environment on learners' behaviour

Both are behavioural theories in which the environment shapes behaviour. A belongs only to classical; B and D only to operant.

23. False

It is the reverse: classical = involuntary, operant = voluntary.

CHAPTER 3 · LESSON 3.4

Schedules of Reinforcement and Shaping

AFTER THIS LESSON YOU CAN

- Distinguish continuous and intermittent reinforcement
- Identify fixed-ratio, variable-ratio, fixed-interval and variable-interval schedules
- Compare schedules for response rate and resistance to extinction
- Explain shaping and classroom applications of operant conditioning

3.4.1 Continuous and intermittent reinforcement

- **Continuous reinforcement:** reinforce **every** correct response. Learning is **fastest** at first, but the behaviour **extinguishes quickly** when reinforcement stops.
- **Intermittent (partial) reinforcement:** reinforce only **some** responses. Learning is slower, but the behaviour is much **more resistant to extinction**. So: to make a learned behaviour **last**, reinforce it **intermittently**.

3.4.2 The four intermittent schedules

F RATIO = NUMBER OF RESPONSES. INTERVAL = TIME.

Schedule	Reinforcement comes after...	Example	Effect
Fixed ratio (FR)	a **fixed number** of responses	50 birr per basket of coffee picked; 1 bonus mark for every 5 correct answers; a bonus for every 100 patients treated; a free flight every 10,000 miles	high rate; a pause just after each reward
Variable ratio (VR)	an **unpredictable number** of responses (varying around an average)	gambling and lottery; a teacher who praises sometimes after 1 correct answer, sometimes after 3; sales where the payment varies	**highest, steadiest** rate; **most resistant** to extinction
Fixed interval (FI)	the first response after a **fixed time**	a monthly or weekly salary; a bonus every three months; pay every Friday	responding increases as the time approaches
Variable interval (VI)	the first response after an **unpredictable time**	**surprise quizzes**; a boss who checks your work at random times; redialling a busy phone	slow, steady rate; resistant to extinction

? SOLVING A SCHEDULE QUESTION IN TWO STEPS (DEBRE BIRHAN 2015)

"Abebe is paid in cash every two weeks, while Bekele's pay depends on the number of books he sells, and the payment varies." Identify both schedules.

Step 1, **ratio or interval?** Does reward depend on **how much** you do (ratio) or **how long** you wait (interval)? Abebe → time → **interval**. Bekele → books sold → **ratio**.

Step 2, **fixed or variable?** Is it **predictable**? Abebe: every two weeks → **fixed**. Bekele: varies → **variable**.

Answer: Abebe = **fixed interval**; Bekele = **variable ratio**.

Watch the trap words: "every", "each", "per" + a number of tasks → **FR**; "monthly", "every Friday" → **FI**; "sometimes", "unpredictable", "surprise" + responses → **VR**; + time → **VI**.

★ WHICH SCHEDULE IS BEST?

- Fastest learning of a **new** response: **continuous**.
- Most **powerful and productive**, most **resistant to extinction**: **variable ratio** (that is why gambling is so hard to stop).
- Variable schedules (**VR and VI**) resist extinction better than fixed ones.

3.4.3 Shaping

D SHAPING

Reinforcing **successive approximations** of a desired behaviour: reward a first step in the right direction, then require responses that are closer and closer to the final goal. Used for behaviours that would almost never appear on their own (teaching a child to use a knife and fork, training animals, building study habits).

◉ SHAPING IN A CLASSROOM

A shy student never speaks in class. The teacher first praises her for **looking up** during discussion, then for **whispering an answer to a classmate**, then for **answering a yes/no question**, and finally for **giving a full answer** aloud. Each step is a successive approximation.

Classroom applications of operant conditioning: arrange rewards (praise, smiles, marks, prizes) for study behaviour; avoid harsh punishment, which conditions dislike of the teacher and subject; use **programmed instruction** (small logical steps with **immediate** feedback at each step); use **systematic desensitization** to reduce learned anxiety (present a weak form of the feared stimulus and gradually increase it); reinforce question-asking and participation for the whole group.

! SCHEDULE TRAPS

- In a **variable interval** schedule, reinforcement depends on **time**, not on a number of responses.
- Continuous reinforcement gives **faster learning** but is **less** resistant to extinction.
- A fixed grade cut-off ("score 95 % to get an A+") requires a fixed amount of performance, so it is treated as **fixed ratio**.

Exam Practice

18 QUESTIONS · LESSON 3.4

1 Arsi University · Final 2013

Mr. X, a resident of Jimma, picks about 10 baskets of coffee a day and receives 50 birr for every basket. His reinforcement schedule is:

A) Fixed ratio
B) Variable ratio
C) Fixed interval
D) Variable interval

2 Arsi University · Final 2013

The manager of Chilalo Food Complex gives workers a bonus every three months. Which schedule is this?

A) Fixed interval
B) Variable interval
C) Fixed ratio
D) Variable ratio

3 Addis Ababa University · Mid 2014

Your psychology instructor praises students sometimes after one correct answer, sometimes after three, and so on. This schedule is:

A) variable ratio
B) fixed ratio
C) fixed interval
D) variable interval

4 Debre Birhan University · Final 2015

Abebe is paid in cash every two weeks, whereas Bekele's pay depends on the number of books he sells and varies. Abebe and Bekele are on **__ and** ____ schedules respectively.

A) fixed ratio; variable interval
B) variable ratio; fixed ratio
C) fixed interval; variable ratio
D) variable interval; fixed ratio

5 Jigjiga University · Mid 2012

Your boss checks your work occasionally, so you never know when the next check will be. This is an example of:

A) Fixed ratio
B) Fixed interval
C) Variable ratio
D) Variable interval

6 Bahir Dar University · Final 2015

In which reinforcement schedule can students NOT predict how much time must pass before reinforcement occurs?

A) Variable interval
B) Fixed ratio
C) Fixed interval
D) Variable ratio

7 Dire Dawa University · Mid 2014

Gambling behaviour is maintained on which schedule?

A) Variable interval
B) Fixed ratio
C) Variable ratio
D) Fixed interval

8 Dilla University · Mid 2015

Which schedule is most powerful in modifying behaviour?

A) Fixed interval
B) Variable interval
C) Variable ratio
D) Fixed ratio

9 Dire Dawa University · Mid 2014

Which schedules are better at resisting extinction?

A) Variable ratio and variable interval
B) Variable ratio and fixed interval
C) Fixed ratio and fixed interval
D) Fixed ratio and variable ratio

10 Arsi University · Final 2014

If a learned behaviour is expected to resist extinction, reinforcement should be given intermittently.

True / False

11 Dire Dawa University · Mid 2014

Continuous reinforcement is more resistant to extinction than partial reinforcement.

True / False

12 Bahir Dar University · Mid 2014

A continuous reinforcement schedule leads to faster learning than an intermittent one.

True / False

13 Wolaita Sodo University · Mid 2014

In a variable interval schedule, reinforcement is given after a variable number of correct responses.

True / False

14 Wachemo University · Mid 2016

Ethiopian Airlines offers a free flight after every 10,000 miles travelled. The airline is using a:

A) fixed-interval schedule
B) variable-ratio schedule
C) variable-interval schedule
D) fixed-ratio schedule

15 Madda Walabu University · Final 2014

Bonsa works about 40 hours a week as a daily labourer. He is always paid on Friday afternoon, and his pay is always the same. This is an example of:

A) Fixed interval
B) Fixed ratio
C) Variable ratio
D) Variable interval

16 Jimma University · Mid 2014

Rewarding successive approximations toward a target behaviour is:

A) Shaping
B) Extinction
C) Positive reinforcement
D) Negative reinforcement

17 Jimma University · Mid 2015

A teacher continually rewards small improvements until a student reaches the desired goal. This is an example of:

A) Latent learning
B) Positive reinforcement
C) Shaping
D) Vicarious reinforcement

18 Arsi University · Final 2013

If we reinforce the desired response every time it occurs, we are using:

A) incremental reinforcement
B) continuous reinforcement
C) intermittent reinforcement
D) contiguity reinforcement

Answers & Explanations

1. **A**

Fixed ratio

Payment follows a fixed number of responses (one basket).

2. **A**

Fixed interval

The reward depends on a fixed amount of time.

3. A

variable ratio

Praise follows an unpredictable number of responses.

4. C

fixed interval; variable ratio

Abebe: fixed time → FI. Bekele: varying number of sales → VR.

5. D

Variable interval

Checks depend on unpredictable time.

6. A

Variable interval

Unpredictable time = variable interval.

7. C

Variable ratio

Wins come after an unpredictable number of bets.

8. C

Variable ratio

Variable ratio produces the highest, steadiest response rate and greatest resistance to extinction.

9. A

Variable ratio and variable interval

Unpredictable (variable) schedules keep people responding longest when rewards stop.

10. True

The change from intermittent reinforcement to none is less noticeable, so responding continues longer.

11. False

Partial (intermittent) reinforcement is more resistant.

12. True

Reinforcing every response makes the connection clearest during acquisition.

13. False

That is variable ratio. Variable interval depends on varying time.

14. D

fixed-ratio schedule

The reward follows a fixed amount of "responding" (miles flown), not a fixed time.

15. A

Fixed interval

Pay comes after a fixed period of time.

16. A

Shaping

That is the definition of shaping.

17. C

Shaping

Rewarding steps that get closer and closer to the goal is shaping (which uses positive reinforcement step by step).

18. B

continuous reinforcement

Every response reinforced = continuous reinforcement.

CHAPTER 3 · LESSON 3.5

Social Learning and Cognitive Learning



AFTER THIS LESSON YOU CAN

- Explain Bandura's observational learning and the three forms of reinforcement
- State the four conditions for modelling in order
- Explain latent learning (Tolman) and insight learning (Köhler)
- Compare all four learning theories and their classroom implications

3.5.1 Social learning (observational) theory

D OBSERVATIONAL LEARNING

According to **Albert Bandura**, much human learning happens by **watching others** (models) and the consequences of their behaviour, without doing the behaviour or being reinforced ourselves. Because it involves thinking about what we observe, it is also called the **social cognitive** approach.

F THREE FORMS OF REINFORCEMENT IN SOCIAL LEARNING

Form	Meaning	Example
Direct	the observer copies the behaviour and is reinforced personally	
Vicarious	the observer sees **someone else** rewarded and copies the behaviour	Hayelom works harder after seeing his classmate Lencho win a foreign scholarship; students study harder when universities post photos of top achievers
Self-reinforcement	the person rewards **himself or herself**, valuing growing competence	a student feels proud and treats herself after finishing a hard chapter

F BANDURA'S FOUR CONDITIONS FOR MODELLING (IN ORDER)

1. **Attention:** the observer must notice the model (the **first** step).
2. **Retention:** the observer must **remember** the behaviour (rehearsal helps).
3. **Motor reproduction:** the observer must be **able to perform** the behaviour; this is where learning is turned into performance. (Small children may not yet be able to copy complex movements.)
4. **Motivation:** the observer must **want** to perform it.

Memory aid: "**A** **R**eal **R**ole **M**odel": Attention, Retention, Reproduction, Motivation.

S BANDURA'S BOBO DOLL EXPERIMENT

Children who watched an adult **hit and kick an inflatable Bobo doll** later attacked the doll in the same way, even copying the adult's words, while children who saw a calm model did not. Children who saw the model **punished** copied less (but could still do it when offered a reward). This showed that **learning can occur without direct reinforcement**, and that what is learned is not always performed.

Classroom implications: students learn by observing; teachers and parents must **model** appropriate behaviour; show students a variety of models to break stereotypes; build **self-efficacy** (belief in one's capability) through success

experiences and watching others succeed; set realistic but challenging expectations; teach **self-regulation**. Modelling can teach a new behaviour **faster** than shaping.

3.5.2 Cognitive learning theory

Cognitive theorists accept conditioning but argue that humans have **attitudes, beliefs and expectations** that affect how they learn. Learning involves **internal mental processes**, not just stimulus–response links.

D LATENT LEARNING (TOLMAN)

Learning that occurs but is **hidden** and **not shown in behaviour** until there is a reason (incentive) to show it.

Tolman and Honzik's rats: Group 1 always found food in the maze and learned it quickly; Group 2 never found food; Group 3 found no food for 10 days, then food on day 11, and by the next day ran the maze as well as Group 1. They had learned a **cognitive map** without reinforcement.

★ INSIGHT LEARNING (KÖHLER; GESTALT)

A **sudden** reorganization of how we perceive a problem, after a period with no apparent progress, followed by an **"aha" experience**. It does not depend on reinforcement or trial and error, and the solution transfers easily to similar problems.

Köhler's chimpanzee, Sultan, suddenly joined two sticks (or stacked boxes) to reach bananas.

◉ LATENT OR INSIGHT?

- You pass a hospital every day without thinking about it. A stranger asks where it is and you cannot explain. One day a friend is injured and you lead him straight there → **latent learning** (the knowledge was there, shown when needed).
- A girl watches her mother cook. When her mother is away, she prepares the family meal → **latent learning**.
- Limat tries to unscramble GYLOPYHCOS for 20 minutes without success; walking to class, "PSYCHOLOGY" suddenly comes to her → **insight learning**.
- Beka struggles with a maths problem, then suddenly sees how to solve it without being taught anything new → **insight**.

F THE FOUR THEORIES COMPARED

Theory	Theorist	Learning happens through...	Key ideas
Classical conditioning	**Pavlov** (and Watson)	**association of two stimuli**	NS, UCS, UCR, CS, CR; involuntary responses
Operant conditioning	**Skinner** (and Thorndike)	**consequences** of voluntary behaviour	reinforcement, punishment, schedules, shaping
Social learning	**Bandura**	**observing models**	vicarious reinforcement; attention, retention, reproduction, motivation
Cognitive learning	**Tolman**, **Köhler**	**mental processes**	latent learning, cognitive maps, insight

SOCIAL AND COGNITIVE TRAPS

- **Vicarious** reinforcement = seeing **others** rewarded; rewarding **yourself** = **self**-reinforcement.
- In social learning, the learner does **not** need to perform the behaviour or be directly reinforced to learn it.
- **Sudden** understanding = **insight**, not latent learning.
- **Hidden** learning shown later when there is an incentive = **latent** learning.
- Acquiring a fear of a light by seeing **someone else** shocked is **social (observational)** learning, not classical conditioning.

Exam Practice

18 QUESTIONS · LESSON 3.5

1 Bahir Dar University · Mid 2014

Abel scored 30/30 on the psychology midterm and the teacher asked the class to applaud him. As a result, a classmate was motivated and scored 50/50 on the final. What kind of reinforcement did the classmate receive?

A) Direct reinforcement
B) Self-reinforcement
C) Vicarious reinforcement
D) Teacher's reinforcement

2 ASTU · Mid 2015

Some Ethiopian universities post photos and grades of high-achieving students so that others learn from them. If other students are motivated by this, which kind of reinforcement is operating?

A) Positive reinforcement
B) Negative reinforcement
C) Self-reinforcement
D) Vicarious reinforcement

3 Jigjiga University · Mid 2012

Addisu, a first-year student, studies very hard because he saw senior students receive rewards from the department for good performance. This illustrates:

A) Observational learning
B) Operant conditioning
C) Latent learning
D) Classical conditioning

4 Jimma University · Mid 2015

Vicarious reinforcement is the rewarding of oneself for one's appropriate behaviour.

True / False

5 Dilla University · Mid 2015

The first step in learning through observation is:

A) Motivation
B) Attention
C) Reproduction
D) Retention

6 Addis Ababa University · Mid 2014

According to Bandura, learned behaviour is transformed into performance at the phase of:

A) reproduction
B) retention
C) attention
D) motivation

7 Jimma University · Mid 2015

In observational learning, the ability to replicate the behaviour the model has demonstrated is called:

A) Motivation
B) Retention
C) Attention
D) Reproduction

8 Haramaya University · Mid 2015

Acquiring a fear of a light because you saw someone else being shocked when it came on is an example of:

A) Social (observational) learning
B) Classical conditioning
C) Instrumental learning
D) Involuntary learning

9 Dire Dawa University · Mid 2014

Which of the following is NOT true about social learning theory?

A) People can learn by observing others
B) The learner must personally perform the behaviour to learn it
C) Cognition plays a role in learning
D) People adjust their behaviour after seeing others' consequences

10 Addis Ababa University · Mid 2014

Limat tried for 20 minutes to unscramble GYLOPYHCOS but failed. While walking to class, the answer "PSYCHOLOGY" suddenly came to her. This is an example of:

A) classical conditioning
B) operant conditioning
C) insight
D) the law of effect

11 Wolaita Sodo University · Mid 2014

You pass a hospital every day on your way to study without thinking about it. When a stranger asks where it is, you cannot describe it. One day a friend is injured, and you take him straight to the hospital. This is:

A) Insight learning
B) Cognitive map only
C) Latent learning
D) Observational learning

12 Bahir Dar University · Mid 2014

A girl often watched her mother prepare food. When her mother was away, the girl managed to prepare the family's food. She learned this skill through:

A) Operant conditioning
B) Latent learning
C) Insight learning
D) Classical conditioning

13 Arba Minch University · Final 2015

Which of the following is true about latent learning?

A) Trial and error is essential for it
B) Without an incentive, a learner may not show what has been learnt
C) Reinforcement is necessary for the learning itself
D) It happens suddenly with an "aha" experience

14 Jimma University · Mid 2015

The sudden understanding of a solution to a problem is called latent learning.

True / False

15 Madda Walabu University · Final 2014

The good feeling called an "aha" experience after solving a problem is related to:

A) classical conditioning
B) operant conditioning
C) latent learning
D) insight learning

16 Wachemo University · Mid 2016

Which learning theory states that learning takes place by modelling or observing others?

A) Cognitive theory
B) Psychoanalytic theory
C) Social cognitive theory
D) Behavioural theory

17 Arba Minch University · Final 2015

Which of the following is true about observational learning?

A) Reinforcement is required for every observation
B) People learn only through direct engagement
C) People can use self-reinforcement to motivate themselves
D) Cognition plays no role

18 Arba Minch University · Final 2016

According to cognitive learning theory, what is the primary focus of learning?

A) Observable changes in behaviour
B) The internal mental processes involved in learning
C) Emotions only
D) Reinforcement schedules

Answers & Explanations

1. C

Vicarious reinforcement

The classmate was influenced by seeing someone else rewarded.

2. D

Vicarious reinforcement

Students are reinforced by observing others' rewards.

3. A

Observational learning

He learned from watching others' behaviour and its consequences.

4. False

That is self-reinforcement. Vicarious reinforcement comes from seeing others rewarded.

5. B

Attention

The order is attention → retention → reproduction → motivation.

6. A

reproduction

Motor reproduction is where the observer actually performs what was learned.

7. D

Reproduction

Motor reproduction = being able to perform the behaviour.

8. A

Social (observational) learning

You were never shocked yourself; you learned by observing another person.

9. B

The learner must personally perform the behaviour to learn it

In observational learning, people can learn without performing the behaviour themselves.

10. C

insight

A sudden solution after a period without progress is insight learning.

11. C

Latent learning

The learning was hidden until there was a reason to use it.

12. B

Latent learning

The skill was learned earlier but not shown until it was needed.

13. B

Without an incentive, a learner may not show what has been learnt

Latent learning stays hidden until conditions (incentives) make it worth showing.

14. False

Sudden understanding is insight learning.

15. D

insight learning

Insight produces a sudden solution and the "aha" feeling.

16. C

Social cognitive theory

Bandura's social cognitive (social learning) theory centres on observation and modelling.

17. C

People can use self-reinforcement to motivate themselves

Bandura names self-reinforcement as one of three forms of reinforcement.

18. B

The internal mental processes involved in learning

Cognitive theorists emphasize thinking, expectations and understanding.

CHAPTER 3 · QUICK REVISION

Learning and Theories of Learning — at a glance

CHAPTER 3 AT A GLANCE

Topic	Must-know points
Learning	relatively permanent change in behaviour from experience or practice; not fatigue, illness, drugs, maturation, reflexes; inferred, not seen
Characteristics	continuous, pervasive, whole person, responsive to incentives, active, purposeful, multifaceted
Classical	Pavlov; NS → CS; UCS → UCR; CS → CR; generalization, discrimination, extinction (CS without UCS), spontaneous recovery
Operant	Skinner; voluntary; positive = add, negative = remove; reinforcement increases, punishment decreases; primary vs secondary; escape vs avoidance
Schedules	continuous (fast learning); FR (number, fixed), VR (number, varies, strongest), FI (time, fixed), VI (time, varies)
Shaping	reinforcing successive approximations
Social	Bandura; direct, vicarious, self-reinforcement; attention → retention → reproduction → motivation
Cognitive	latent (Tolman, hidden until needed); insight (Köhler, sudden, "aha")

REINFORCEMENT OR PUNISHMENT IN FIVE SECONDS

Ask "**Will the behaviour go up or down?**" (up = reinforcement, down = punishment), then "**Was something added or taken away?**" (added = positive, taken away = negative).